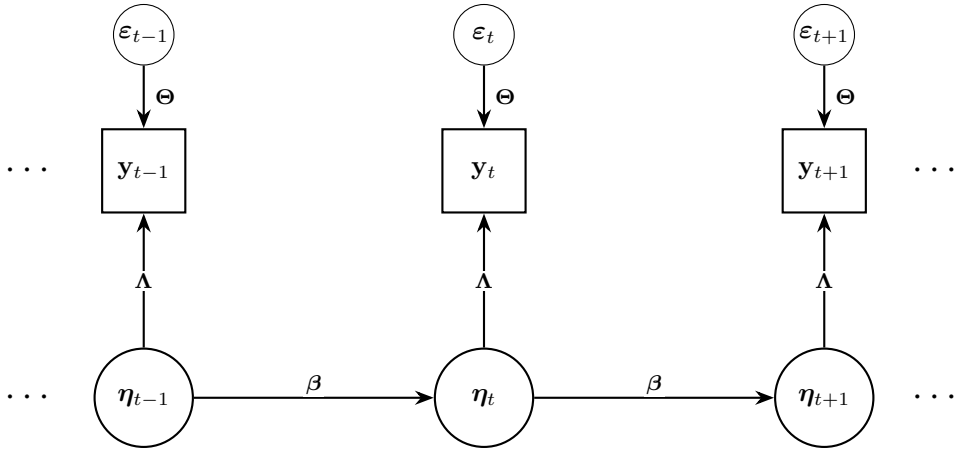


$$\boldsymbol{\beta} = \begin{pmatrix} 1+b_1 & g_{12} & 1 & 0 \\ g_{21} & 1+b_2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\boldsymbol{\Theta} = \begin{pmatrix} \theta_{11} & 0 \\ 0 & \theta_{22} \end{pmatrix}$$

$\boldsymbol{\eta}_t = (\eta_1, \eta_2, \eta_3, \eta_4)'$ (true levels η_1, η_2; slopes η_3, η_4), $\quad \mathbf{y}_t = (Y_1, Y_2)'$; no process noise